

Task 1:

Key Differences Between Ethernet and Fiber Optic Cables

Objective: Compare Ethernet (copper) cables and Fiber Optic cables based on transmission medium, speed/distance, and interference resistance, providing real-world application examples for each.

1. 3 Key Differences Table:

Feature / Criteria	Ethernet Cables (e.g., Cat6 / Copper)	Fiber Optic Cables
1. Transmission Medium	Transmits data using electrical signals over copper wires.	Transmits data using light pulses through thin glass/plastic strands.

Feature / Criteria	Ethernet Cables (e.g., Cat6 / Copper)	Fiber Optic Cables
2. Speed & Maximum Distance	Supports speeds up to 1–10 Gbps with a maximum effective length of 100 meters (328 ft).	Supports ultra-high speeds (100+ Gbps) over long distances (several kilometers) without signal loss.
3. EMI Resistance	Susceptible to Electromagnetic Interference (EMI) from power lines and motors.	Completely immune to EMI and electrical noise.

2. Real-World Application Examples:

- **Ethernet Cable Real-World Use Case:**

- **Example:** Connecting a desktop PC or Smart TV to a **Home Wi-Fi Router** inside a room.
- **Why:** Short distances (<100m) make Cat6 Ethernet cables cost-effective, easy to crimp, and ideal for local home networks.

- **Fiber Optic Cable Real-World Use Case:**

- **Example: College Campus Backbone** or Inter-building ISP infrastructure.
- **Why:** High bandwidth and long-distance capability allow connecting multiple distant campus buildings together without signal degradation or electrical interference.

Task 2:

T568B Straight-Through Ethernet Cable Wiring Diagram

Objective: Document and diagram the pinout configuration and wire color standards for a straight-through Ethernet cable using the **T568B wiring standard**.

1. T568B Pinout Configuration Table:

Pin Number	Wire Color Standard (T568B)	Function / Signal
Pin 1	White / Orange	Transmit Data + (TX+)
Pin 2	Orange	Transmit Data - (TX-)
Pin 3	White / Green	Receive Data + (RX+)
Pin 4	Blue	Unused (Power / Reserve)

Pin Number	Wire Color Standard (T568B)	Function / Signal
Pin 5	White / Blue	Unused (Power / Reserve)
Pin 6	Green	Receive Data - (RX-)
Pin 7	White / Brown	Unused (Power / Reserve)
Pin 8	Brown	Unused (Power / Reserve)

Task 3:

Step-by-Step Guide to Crimping an RJ45 Connector on a CAT6 Cable

Objective: Document a step-by-step procedure for terminating a CAT6 Ethernet cable with an RJ45 connector using standard cable crimping tools and the T568B wiring scheme.

1. Tools Required:

- **CAT6 Ethernet Cable**
 - **RJ45 Connectors** (CAT6 compatible)
 - **RJ45 Crimping Tool**
 - **Wire Stripper / Cable Jacket Cutter**
 - **Network Cable Tester**
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2. Step-by-Step Termination Procedure:

1. Strip the Outer Cable Jacket:

- Measure about 1 to 1.5 inches from the end of the CAT6 cable.
- Insert the cable into the wire stripper, rotate it around once, and gently pull off the outer PVC jacket without damaging the inner twisted pairs.

2. Cut the Plastic Center Spine:

- CAT6 cables contain a central plastic core/spine separating the pairs. Pull back the 4 twisted pairs and cut the center spine flush with the outer jacket using wire cutters.

3. Untwist and Straighten Wires:

- Untwist all 4 pair combinations (8 individual wires) and flatten/straighten them out completely using your fingers.

4. Arrange Wires according to T568B Standard:

- Align the 8 wires side-by-side in the correct **T568B color sequence** from left to right:

1. *White-Orange*
2. *Orange*
3. *White-Green*
4. *Blue*
5. *White-Blue*
6. *Green*
7. *White-Brown*
8. *Brown*

5. Trim Wires to Even Length:

- Pinch the arranged wires tightly between your fingers. Using the wire-cutting blade on the crimping tool, make a clean, straight flush cut across the top, leaving about **0.5 inches (1.2 cm)** of exposed wire.

6. Insert Wires into the RJ45 Connector:

- Hold the RJ45 connector with the **locking plastic clip facing DOWN** (copper pins facing UP).
- Push the flattened wires firmly into the connector until all 8 conductors reach the top end of the connector pins and the outer cable

jacket enters the rear sleeve of the connector for strain relief.

7. Crimp the RJ45 Connector:

- Insert the loaded RJ45 connector into the RJ45 slot of the crimping tool.
- Squeeze the crimper handles firmly together until you hear/feel the metal pins puncture the individual wire insulation and secure the locking tab.

8. Test the Connection:

- Repeat the process on the other end of the cable.
- Plug both ends into a **LAN Cable Tester**. Power ON the tester and verify that all lights (**1 through 8**) flash green sequentially, confirming a fully functional straight-through cable.

Task 4:

Gaming Café Cable and Tool Selection Scenario

Objective: Select the appropriate network cable type and termination tool for a 10-PC LAN gaming café based on speed, distance, and budget factors.

1. Selections Made:

- **Cable Type Selected: CAT6 Ethernet Cable (Copper)**
 - **Tool Selected: RJ45 Crimping Tool**
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2. Justification (3-4 Sentences):

For setting up a 10-PC LAN gaming café, **CAT6 Ethernet cables** are the ideal choice because the total wiring distance within the room is well under 100 meters, and CAT6 easily supports 1 Gbps to 10 Gbps speeds with extremely low latency for seamless multiplayer gaming. Utilizing an **RJ45 crimping tool** allows quick, cost-effective, and custom-length cable terminations without requiring expensive specialized equipment. In contrast, Fiber Optic cables and Fiber Cleavers are overly expensive and unnecessary for short-range local area network (LAN) environments.